

# Mahad Hassan

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## EDUCATION

### McMaster University

Honours Software Engineering Co-op Level III, B.Eng (3.9 GPA)

Expected May 2027

Hamilton, ON

- **Achievements:** Consistent Dean's List | Finalist at MacEngComp 23' | \$3k entrance scholarship
- **Coursework:** Data Structures & Algorithms | OOP (Java) | Development (C, Git, Bash) | Databases (SQL)
- **Leadership:** Events VP, MacPSA | VP Operations, Voices@Mac | Attendee Relations, DeltaHacks | Web Dev, MacSES

## TECHNICAL SKILLS

**Languages:** Python, Java, C/C++, TypeScript/JavaScript, SQL, Bash, MATLAB, Verilog, HTML/CSS

**AI & Agents:** LLM Agents, MCP Servers (FastMCP), OpenAI SDK, Agent Skills, RAG (LlamaIndex), LLM Fine-tuning (Unsloth), Prompt/Context Engineering

**Testing & Automation:** Jenkins, Hardware-in-the-loop Validation, Test Frameworks, Unit/Integration Testing, Log Analysis

**Machine Learning & CV:** TensorFlow, OpenCV, NumPy, Pandas, Matplotlib, Synthetic Data, Classical CV

**Simulation & Robotics:** Sensor Modeling, Domain Randomization, Sim-to-Real Transfer, Control-Loop Testing

**Embedded & Systems:** Linux/UNIX, Jetson Nano, Raspberry Pi, Firmware (C/C++), Telemetry, Serial I/O

**Web/App:** FastAPI, React, Next.js, Astro, Node.js, React Native, Flask, REST APIs

**DevOps & Cloud:** Git/GitHub, GitHub Actions, CI/CD, Docker/Compose, Helm/Kubernetes, Terraform, Kafka, Azure/AWS

**Data:** PostgreSQL/SQL, SQLAlchemy, Firebase/NoSQL, JSON Schema, YAML

**Practices:** Agile/SDLC, Jira, Confluence, Code Reviews, Release/Incident Support

## EXPERIENCE

### Software Test Automation Engineer Intern

May 2026 – Present

Advanced Micro Devices (AMD)

Markham, ON

- Built the **Test Content Agent**, a **fully autonomous** pipeline that gathers requirements, generates tests, and validates them on real hardware with **no human in the loop** until review, cutting generation from ~1 week to hours (~99%) across **48** generated tests
- Built the entire **UI** for tracking and monitoring the pipeline, plus most of the backend on **FastAPI** and **PostgreSQL**, orchestrating LLMs and pulling requirements from **5 sources** including Jama, Confluence, and internal test storage sites
- Agents run **50–90 searches** per full run across sources including an **845k-line, 2,094-test** library, reading **1.5–5M tokens** and condensing it into ~10k lines of research
- Designed its **self-heal loop** with **rollback**, catching ~91% of false passes from tests that never actually ran
- **Containerized** it with Docker for an **org-wide Kubernetes** rollout, with a watchdog for recurring pipeline failures
- Built an **MCP server** that lets agents **autonomously** write, run, and debug tests on real hardware via **Jenkins**, with SSH validation and **log compression** saving **3 hours** per run
- Building **Atlas**, a mono-repo codifying org-wide engineering knowledge into structured, agent-readable form with validators and schema-based governance, so agents retrieve only the context a task needs, as the foundation for **org-wide** knowledge
- Wrote reusable agent skills and PR workflows for Atlas; also migrated MS Teams and AgilitySDK tests

### Software Engineer Intern

May 2025 – Aug 2025

TD Bank

Toronto, ON

- Developed **FRAM**, a Java/Spring microservice on Azure using **Kafka** for high-volume asynchronous processing
- Migrated API client processing to **non-blocking, concurrent** execution, delivering **\$1M/year** in infrastructure savings
- Analyzed performance and failure modes across distributed components to improve reliability under production load
- Improved **CI/CD** via GitHub Actions for **20% faster deployments** across **7 microservices**
- Updated **Terraform** configurations to enable **failover testing** in secondary environments
- Wrote unit/integration tests and supported releases/incidents as part of the team's SDLC

### Machine Learning Engineer

Oct 2025 – Present

McMaster Aerial Robotics and Drones Club

Hamilton, ON

- Implementing a real-time **vision model pipeline** using TensorFlow + classical CV for autonomous landing
- Optimizing ML inference on the **Jetson Nano** (CUDA-accelerated) for **sub-50ms**, real-time processing
- Iterating on preprocessing, input resolution, and model parameters to balance accuracy with real-time constraints
- Integrating detection outputs with **precision-landing controllers** and waypoint-navigation logic
- Applying **domain randomization** (lighting, noise, textures, occlusions) to improve robustness and sim-to-real transfer

- Fusing RGB, depth, and telemetry streams to stabilize predictions and increase landing accuracy at **4–5 m**

## Website Developer

Jul 2025 – Present

McMaster Software Engineering Society

Hamilton, ON

- Build the **SES website** using **Astro + TypeScript**, improving navigation and discoverability for **500+ students**
- Develop features (e.g., merch listings) via PRs and reviews, improving delivery speed by **30%**
- Structure reusable components and documentation to reduce feature development time by **25%**

## App Developer

Jan 2025 – Present

McMaster iBioMed Society

Remote

- Develop a **React Native + Firebase** app centralizing mental, physical, and financial support resources for patients
- Implement end-to-end **encryption**, a secure medical resume section, and protected data storage
- Design an accessible, privacy-first UX for secure and seamless patient interaction

## Controls Subteam Member

Sep 2025 – May 2026

McMaster Rocketry

Hamilton, ON

- Built **MATLAB/Simulink** simulations of flight dynamics and sensor behavior to test control loops
- Implemented **C++ firmware** for IMU filtering, telemetry ingestion, and actuator response on flight hardware
- Developed a **TypeScript** dashboard to visualize real-time strain-gauge data and validate loads against simulations
- Built tools to **view, record, and trim** telemetry logs; maintained a modular codebase for repeatable workflows

## PROJECTS

### PrepPal | *TypeScript (Next.js/React), GCP Vision, OpenAI*

**3rd Place @ MacEngComp 25'**

- Developed an **AI-powered** disaster-readiness platform using **Next.js + GCP Vision** with **<2s** image detection
- Designed a modular detection pipeline (**localization + OCR + label matching**) improving reliability by **40%**
- Integrated **global hazard data** (USGS/NOAA/NASA) enabling **instant (<1s)** risk lookups and threat analysis
- Handled noisy, real-world image inputs by combining model predictions with rule-based validation for safer outputs

### SecureVault | *Python (Flask/OpenCV/Cryptography), SQL*

**2nd Place @ MacEngComp 24'**

- Designed a system combining **facial recognition, password manager, and file encryption** to enhance data protection
- Leveraged Python and SQL to develop a solution within a **7-hour coding sprint**, securing **2nd place** among 30+ teams
- Focused on secure handling of biometric data and encrypted storage under strict time constraints

### Digit Recognizer AI | *TypeScript (React), Python (TensorFlow)*

- Designed an **8-layer CNN** achieving **99.3%** accuracy on MNIST using **60,000** training and **10,000** test images
- Utilized **Matplotlib** to visualize model predictions with confidence levels for each digit class, including probabilities
- Integrated **React & TailwindCSS** to visually display **live** bar chart predictions for drawn digits
- Experimented with network depth and pooling strategies to study their impact on convergence and generalization
- Used **data augmentation** (shifts, rotations, noise) to test model robustness under domain variation

### C-View | *C, Bash, Git*

- Developed a **C-based image editor** to apply filters like **grayscale, reflection, rotation, edge detection, and blur**
- Designed to process images up to **30%** faster than comparable tools with reduced memory use